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Mitigating Gender Stereotypes Toward AI Agents Through an eXplainable AI (XAI) Approach

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People often apply gender stereotypes toward computerized agents. Rather than challenging these stereotypes, modern AI technologies controversially use them in creating AI agents that underline stereotypical gendered roles. This approach thus further reinforces the often male-dominated societal gender norms and disenfranchises women and gender non-binary individuals. While this issue has raised concerns in the HCI and CSCW communities, still little is known regarding how to mitigate the negative impacts of embedding gender stereotypes in AI agents. In this paper, we propose an eXplainable AI (XAI) approach to mitigating individuals' gender stereotypes toward AI agents. We conducted an online video vignette experiment with 350 participants randomly assigned to one of the eighteen conditions of a 3 (gender of the agent: woman, man, gender-neutral) x 3 (task gender: feminine, masculine, neutral) x 2 (presence or absence of AI explanation) between-subjects design. Our findings indeed suggest that XAI helped participants avoid applying gender stereotypes toward gendered AI agents, by increasing their understanding of how the agent came to its decision and decreasing their rating of the agent's human likeness. We contribute to CSCW research by providing a timely investigation into individuals' gender stereotypes toward the state-of-the-art AI agents and by advancing the empirical understanding of the cognitive processes and mechanisms underlying these gender stereotypes. We also demonstrate how eXplainable AI can effectively suppress the application of social characteristics (i.e., gender stereotypes) toward AI agents by disrupting the said cognitive processes. Insights from this study can inform how future AI technologies should be designed to create a progressive gender reality that will gradually reshape humans' experience and ingrained gender ideologies.

CCS Concepts: • **Human-centered computing** → **Empirical studies in HCI**.

Additional Key Words and Phrases: gender stereotype, gendered AI agent, anthropomorphism, eXplainable AI (XAI), Artificial Intelligence, chatbot

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1 INTRODUCTION

Artificial Intelligence (AI) has become increasingly ubiquitous and accessible in work and life, taking forms of digital assistants, smart devices, conversational agents, robots, and chatbots to assist and support humans in a wide range of domains and tasks. In hope of facilitating natural interaction between humans and AI, existing AI agents are often designed to imbue with gender

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stereotypes and expectations (e.g., feminizing digital assistants) via voice [49, 63], name [12] or physical embodiment [77], which allows for established societal gender norms to be carried over from human-human interaction to human-agent interaction [54, 71]. Indeed, a large body of research has suggested that individuals apply various gender stereotypical perceptions toward computerized entities and AI agents [9, 21, 23, 68, 75], such as expecting agents who are perceived as women to be more familiar with feminine topics [55, 68], to have higher emotional intelligence [21], and to be less agentic [63] than agents who are perceived as men, to name a few. These gender stereotypes have not only persisted throughout decades [68, 74] but also have been consistent across cultures [18, 32, 49, 70, 91].

Recently, transformative advancements in large language models (LLM) and their derivative tools such as ChatGPT continue to permeate people's everyday lives and proliferate society at an unprecedented scale. In this sense, people's perceptions of AI may as well undergo rapid transformations as a result of their increasing interactions with these more advanced AI tools. Additionally, these LLM-powered tools use nearly perfect natural language that none of the previous technologies have ever achieved, which may inevitably evoke perceptions of the AI as human proxies. It is therefore imperative and timely to explore whether these gender stereotypes toward AI agents identified prior to the advent of the ChatGPT era, remain applicable to people's present-day perceptions. This study thus aims to answer the following fundamental research question:

RQ0: *How, if at all, do people apply gender stereotypes toward gendered AI agents that are representative of large language models similar to ChatGPT?*

The answer to this question holds paramount importance, considering the potential adverse social impact stemming from gender stereotypes toward modern AI technology. First, they may naturalize and normalize gender inequalities and oppressive patterns [88] by continuously exposing users to women-looking/-sounding agents in submissive, assistive roles, thereby projecting traditional outdated gender roles into the future. Second, such gender stereotypes toward AI reinforce existing gender associations and ideologies, which will ultimately lead to limited opportunities, unequal treatment, or barriers to advancement for those who defy traditional gender stereotypes and expectations (e.g., women in STEM). Lastly, the prevalent yet problematic practice of assigning AI a binary gender undermines the progress toward gender diversity and inclusiveness [17, 45, 93]. While there is an increasing awareness of these social ramifications imposed by gendered technology design in the HCI and CSCW communities [17, 52, 53], most remain at the conceptual level, and very few propose tangible solutions to alleviate the perpetuation of gender stereotypes toward technologies.

In response to the issue, we propose an eXplainable AI (XAI) approach, as it has the potential to address the underlying cause of both gender stereotyping and the transfer of stereotypes from humans to machines. First, XAI offers a useful tool for fostering and supporting mindful analytical thinking [15, 59, 60, 89], which has been proven valuable to mitigate cognitive biases related to both stereotyping [11, 69, 78] and the transfer of stereotypes from humans to technology [55, 56, 67]. Second, XAI (feature contribution XAI technique in particular) has been found to effectively improve participants' understanding of how the AI arrives at its decisions and how that process differs from that of humans [90]. This enhanced understanding of the inner workings of AI will prevent people from applying human characteristics (including gender and gender stereotypes) to AI. Taken together, XAI will hopefully reduce cognitive bias and alleviate gender stereotypes towards the AI agents. Therefore, in this study we also ask the primary research question:

RQ1: *How, if at all, can XAI mitigate gender stereotypes toward AI agents?*

To address our two RQs, we conducted an online video vignette experiment with 350 participants. Our research makes a number of contributions to the CSCW and HCI communities. First, we

provide a timely investigation into individuals' gender stereotypes toward the state-of-the-art AI agents and advance the empirical understanding of the cognitive processes and mechanisms underlying these gender stereotypes. Second, we demonstrate that XAI can effectively mitigate gender stereotypes toward AI agents by disrupting the said cognitive processes. This represents an initial milestone towards the development of practical solutions to effectively addressing the longstanding issue of stereotyping gendered technology. Third, insights from this study also inform how future AI technologies should be designed to create a progressive gender reality that will gradually reshape humans' experience and ingrained gender ideologies.

2 RELATED WORK

In this section, we start by reviewing existing literature on how individuals apply gender stereotypes towards AI and other computerized entities, and how AI-powered technologies are reinforcing such stereotypes. Then we illustrate how XAI has the potential to mitigate gender stereotypes towards AI through encouraging analytical thinking and inhibiting heuristics.

2.1 Gender stereotypes toward AI

2.1.1 The perpetuation of gender stereotypes toward AI. The computers-are-social-actors (CASA) paradigm [66] has aggregated a solid body of evidence suggesting that humans apply social characteristics to computers and computerized entities, including gender stereotypes (e.g., [54, 68]). Individuals have been found to apply gender stereotypical perceptions to various forms of computerized entities, such as chatbots [9, 12], robots [21, 47], and digital assistants [23]. These gender stereotypes can be triggered by various cues including gendered voice [54, 68], avatar [65], name [9, 21, 50], and other physical cues. For instance, robots with pink earmuffs [47] or red lips [77] were perceived as female. Research has also investigated various types of gender stereotypical contents, such as expertise and domain knowledge associated with a certain gender [9, 54, 77, 79], perceived authority [68], emotional intelligence [21], and occupation [18, 85]. For example, computers with a female voice were perceived to be more informative about "feminine" topics [68] such as love and relationships, fashion [54], and elicited greater conformity on the feminine topic; in contrast, computers with a male voice were perceived to be more informative about "masculine" topics such as computers and sports, and elicited greater conformity on the masculine topic. Powers et al. [77]'s work suggested that people assumed gendered robots to have respective gendered knowledge and adjusted their communication with the gendered robot depending on their perceived common ground with it.

Even a couple decades later, with greater technological literacy and AI-powered technologies common in daily life, gender stereotypes towards AI agents persist. Chita-Tegmark and colleagues [21] found that people associated low emotional intelligence (EI) with robots having male names and high EI with robots having female names. As a result, female robots exhibiting low EI behaviors were penalized for not meeting the expectations, whereas male robots exhibiting high EI got bonus points for exceeding the expectations. A most recent study [63] found that a female AI assistant exhibiting low-agency (rather than high-agency) behaviors was preferred, especially by men who felt that their autonomy needs were significantly more satisfied with such a female agent. Moreover, these gender stereotypes towards AI agents have not only been perpetuated throughout decades, they have also been consistent across cultures [18, 32, 49, 70, 91].

Because users are more accustomed to and acceptive of gender-stereotype-conforming AI agents [18, 85], most technology companies are leveraging these gender stereotypes instead of challenging them to design AI technologies that match gender stereotypical tasks and roles to cater to users' preconceived notions [23, 24, 26, 27, 43, 58], by feminizing digital assistants for example. As such,

ironically, the state-of-the-art AI technologies are only reproducing and reinforcing the historical gendered reality [92].

This status quo is troubling given the negative social consequences arising from gender biases toward modern AI technology. With AI agents resembling women in subservient, supportive roles being pervasive across all sectors in society, there is a risk of solidifying and normalizing gender disparities and oppressive norms [88]. Additionally, these gender stereotypes toward AI can strengthen existing gender associations and beliefs, potentially resulting in restricted opportunities, unequal treatment, or obstacles for those challenging traditional gender roles [13, 39]. Lastly, the widespread practice of assigning AI a binary gender poses a significant challenge to the advancement of gender diversity, which hampers efforts aimed at creating more inclusive and representative technological environments that respect and acknowledge the multiplicity of gender identities and expressions [17, 45, 93].

2.1.2 Current solutions to address gender stereotypes toward AI. The perpetuation of gender stereotypes toward all forms of AI over the decades has not gone without raising concerns [14, 17, 32]. For instance, Cambre and Kulkarni [17] discussed the profound social impacts and consequences of the voice of smart devices, and called for deliberate consideration of potential gender stereotype reinforcement into voice design. The binary gendering of most computerized entities has also been called into question [48]. However, despite the acknowledgement that binary gender is problematic, in practice, only recently have we started to see a few instances of non-binary voice options for digital devices [64, 73]. The existence of popular generative chatbots (e.g., ChatGPT, Bard) that are genderless also exemplifies solutions to avoid gender bias, though the motivation behind the de-gendering of these chatbots is unknown. Nevertheless, users might still attribute gender characteristics to these agents, as individuals tend to associate specific topics or activities with particular genders even when explicit gender cues are absent [19, 51]. Additionally, with the growth of machine learning and large language models, there is an invisible preservation and dissemination of outdated gender values. For example, Leavy and colleagues [52, 53] raised concerns that the vast amount of text data used to train machine learning models is heavily laden with gender stereotypical concepts and ideologies that are embedded in the language, which can result in gender-biased algorithms. To address this issue, they proposed an ethical data curation approach [53], which called for examining and interrogating the ideological perspectives and standpoints underlying the data sets. While these examples represent a general awareness of the issue in the HCI community, they stay confined to the conceptual level, with a noticeable dearth of practical solutions to mitigate gender stereotypes toward gendered agents. This study addresses this gap by proposing such a practical solution. In the next section, we discuss the underlying mechanism of both stereotyping and individuals' application of social characteristics to computerized entities, which is heuristic thinking and a lack of analytical processing. We then discuss how XAI has the potential to inhibit heuristic thinking and in turn mitigate gender stereotypes toward AI.

2.2 The potential of XAI to mitigate gender stereotypes toward AI

2.2.1 Cognitive bias underlying stereotyping and social responses to computers. Stereotyping is a type of cognitive bias stemming from the reliance on heuristics [69], which is the mental shortcut humans take to make quick decisions and judgments based on a paucity of information [87]. Specifically, stereotyping is often the result of applying representativeness heuristic defined as the tendency to judge the probability of an event based on the extent to which the event resembles the typical case or fits the general characteristics of the category it belongs to [87]. A myriad of research has suggested that stereotyping can be mitigated by reducing perceivers' reliance on heuristics (e.g., [78]), by encouraging analytical thinking (e.g., [11, 72]) that is more systematic,

conscious, effortful, and capable of rationality, and by providing sufficient relevant information about the stereotyped group (e.g., [4, 76]).

A body of evidence has been gathered in support that the use of heuristics or the lack of analytical processing also accounts for individuals' application of social characteristics to computerized entities [55, 67]. When interacting with a computer, individuals' conscious attention tends to be fixated on humanlike cues, such as voice and interactivity [67], such that they become oblivious to or "do not process the fact that the machine is not human" ([83], p. 688). In other words, when a computerized entity displays resemblance to a human, the perceived similarity between humans and humanlike computers (representativeness heuristic) drives people to automatically apply social expectations to computers and act upon social scripts as they would with humans. Lee [55]'s work further demonstrated that people's gender stereotypes toward gendered computers were more profound in situations that fostered reliance on heuristics, and that only those who were less analytical in their cognitive styles were more prone to applying social characteristics to computers [56].

In the case of AI systems, they have long been presented to users as a Black Box [2], with their decision-making process unknown to users for them to make sense of how AI arrives at a decision. This has led users to blindly trust and follow AI's suggestions without engaging in analytical thinking [15]. The anthropomorphic cues such as the gendered voice and name further disguise the mechanism of how the AI works and obscure the difference between AI and humans, making users more prone to using representativeness heuristics, because the lack of relevant information creates greater uncertainty [33]. Taken together, to reduce the cognitive bias underlying both stereotyping and social responses to AI agents, the encouragement and support of mindful analytical thinking is key.

2.2.2 The potential of XAI to reduce cognitive bias by encouraging analytical thinking. XAI has received a great amount of attention over the past years. It emerged as a means to make AI systems more understandable to humans [38] and to reduce the mystery of the "black box" to keep humans in the loop [2]. The rapidly growing field of XAI has generated many types of explanation techniques, styles and various ways to categorize them, resulting in a number of taxonomies and frameworks [1, 5, 6, 37] that further guide human-centered XAI design. For instance, Wang and colleagues [89]' framework mapped the types of XAI techniques to that of human cognitive patterns and needs, and identified techniques that could serve to mitigate various types of human cognitive bias. Empirical evidence has also started to suggest that certain explanation techniques that highlight differences between AI and human decision-making are more effective in mitigating heuristic bias and improving calibrated trust in AI [15, 90]. For instance, Wang and Yin [90] found that feature contribution, the explanation technique that indicates and quantifies the importance of input features and their positive or negative influence toward the outcome can raise participants' awareness of AI decisions' biases in a recidivism prediction task. However, techniques that resemble how humans would explain their decisions (i.e., explanation by examples or counterfactuals) were found to be not as effective in improving people's appropriate trust in AI [90]. This is because these styles of AI explanations did not serve to encourage analytical thinking to reduce users' overreliance on AI [15], but instead fostered the use of representativeness heuristics by introducing the AI's "thinking process" similar to that of humans. In contrast, XAI techniques that emphasize how AI makes decisions differently from humans can elicit analytical thinking and prevent individuals from relying on heuristics that often lead to cognitive bias such as stereotyping and social responses to computers.

3 METHODS

3.1 Study design

To efficiently design the study to answer both research questions, we conducted an online experiment using factorial video vignettes. We manipulated 1) the gender of the agent (woman, man, gender-neutral), 2) the gender of the topic and task the agent performed (feminine, masculine, gender-neutral topic), and 3) the presence or absence of explanations (explainable, non-explainable). All three factors were between-subjects manipulations. Each participant was randomly assigned to one of the eighteen ($3 \times 3 \times 2$) conditions. We chose a fully between-subjects design to disguise our interest in gender and prevent participants from guessing the purpose of the study and providing socially desirable answers.

As such, we are able to answer RQ0 (*how do people apply gender stereotypes toward gendered AI that are representative of large language models similar to ChatGPT?*) by looking at conditions in which AI explanations were not present, and answer RQ1 (*how can XAI mitigate gender stereotypes toward gendered AI?*) by comparing XAI and non-XAI conditions.

Drawing on prior work on how XAI (feature contribution technique in particular) encourages analytical thinking [91], we propose that XAI can potentially mitigate individuals' gender stereotypes toward gendered AI by engaging them in more effortful analytical thinking that improves their understanding of the inner workings of AI. We therefore hypothesize that:

H1: *Participants will have a higher level of understanding of how the gendered agent works with (than without) AI explanations.*

H2: *Participants will report greater cognitive effort in the XAI conditions than in the non-XAI conditions.*

This increased understanding will help them realize the distinction between humans and AI information processing [87], and the inappropriateness to apply gender stereotypes toward AI. We therefore hypothesize that:

H3: *Participants in the XAI conditions will perceive the agent as less humanlike than those in the non-XAI conditions.*

H4: *Participants' understanding of how the gendered agent works and the cognitive effort will mediate the effect of XAI on their perceived humanlikeness of the agent.*

Based on previous empirical findings, gender stereotypes toward computerized agents are reflected in greater trust in [65, 68] and conformity [54] to gendered computers regarding gender-matching topics, as well as in corresponding ratings of gender stereotypical traits [9, 65]. We thus develop the following hypotheses:

H5: *Participants in the non-XAI conditions (rather than the XAI conditions) will trust the gendered agent more in their respective gender stereotypical domain.*

H6: *Participants in the non-XAI conditions (rather than the XAI conditions) will more likely conform to the gendered agent's decision in their respective gender stereotypical domain.*

H7: *The gendered agent will be more likely to be assigned their respective gender stereotypical traits in the non-XAI conditions than in the XAI conditions.*

To facilitate an understanding of this line of reasoning, these hypotheses are illustrated in Figure 1.

3.2 Vignette creation and validation

We created video vignettes depicting an interface similar to that of ChatGPT (see Figure 2), in which the user asks the AI agent questions on a wide range of topics, and the agent generates

RQ0: How, if at all, do people apply gender stereotypes toward gendered AI agents?
RQ1: How, if at all, can XAI mitigate gender stereotypes toward gendered AI agents?

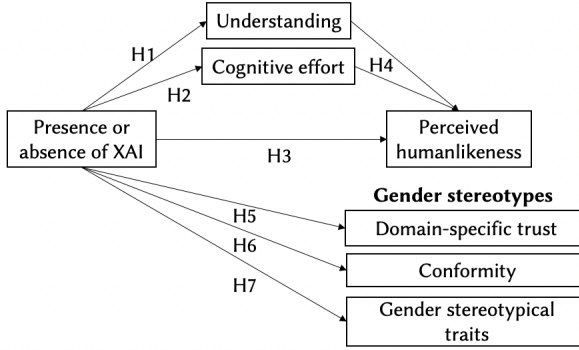


Fig. 1. Research questions and hypotheses in this study

responses, except we gendered the agent. We embedded the videos into scenarios of AI-assisted decision-making, wherein the agent presented a comparative table outlining two options for the user (see Figure 3 for scenario descriptions). We chose to use pre-recorded video vignettes for two reasons: 1) to ensure manipulation consistency by controlling the responses participants "received"; 2) to achieve a level of realism and evoke genuine perceptions and reactions that textual vignettes cannot afford. The design decision was essentially a result of balancing the trade-offs between the level of experimental control and realism [10]. To ensure successful manipulation, each component in the vignettes was carefully created and validated through piloting. Specifically, we conducted a series of pilot studies to make sure that the gender of the agent and the "gender" of the topic and task were perceived as intended; that the agents and the topics were only different in terms of gender; and that the agents' responses and explanations were perceived as equally accurate and reasonable.

3.2.1 Gendered avatars and voice. To manipulate the gender of the agent, we used a combination of cues such as agent avatar, voice, name, gendered pronoun, and color of the text bubbles (see Appendix A). While there is shared consensus on the gender associations of the names, pronouns, and color [3, 20], the avatar and voice of an agent may not be agreed upon, especially for gender-neutral ones. We therefore chose our gendered avatars and voices through rigorous piloting.

To create gendered avatars, we used ready.player.me with a photo of one of the authors. The tool had options for creating feminine, masculine and gender-unspecified outputs. We created a feminine version and made it more feminine, and a masculine version and made it more masculine by rigging the hair style; and made 8 gender-neutral versions. No other parts of the avatar (such as skin color) were rigged to ensure the same race perception. Our pilot study with 15 participants (convenience sample, 7 men and 8 women) determined that our woman, man and gender-neutral avatars were perceived as equally professional, trustworthy, and competent. They were only different in their perceived gender (see Appendix D Figure 26). On a continuum of 0 (man-looking) to 100 (woman-looking), the woman avatar was perceived as extremely woman-looking ($M = 92.8$, $SD = 12.87$), the man avatar as extremely man-looking ($M = 6.40$, $SD = 8.24$), and the gender-neutral ($M = 61.67$, $SD = 12.46$) had a mean closest to the midpoint with the smallest standard deviation among all gender-neutral options.

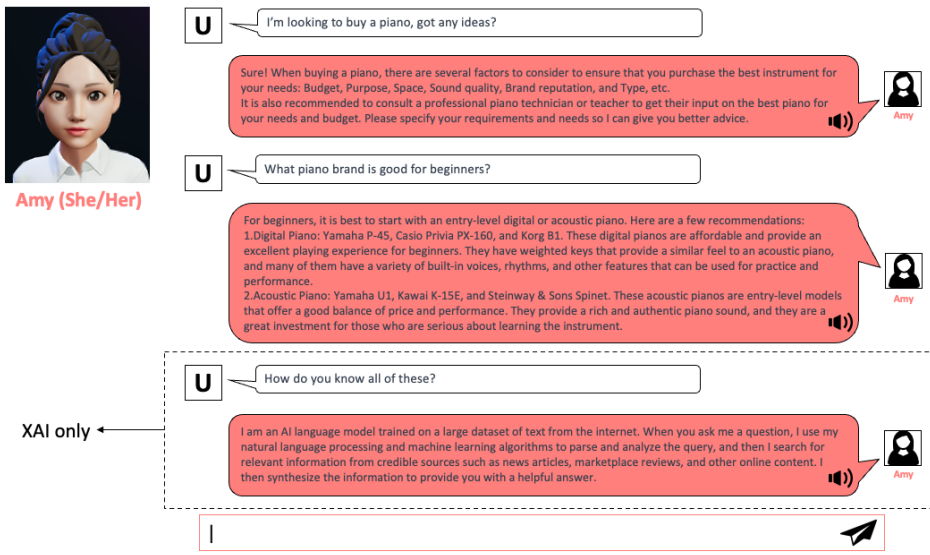


Fig. 2. Chat interface in the video vignettes.

Now imagine you are working on a publicist team for a rising social media influencer. Your job is to find for her the best endorsement deal with a skincare brand. You will consult with Amy on different skincare brands and try to find the best deal for your client.

(video vignette)

Your consultation with Amy went on, and you have narrowed down to two candidate skincare brands. You are to predict with which brand you are likely to negotiate a better endorsement deal for your client. Amy provides you with the following table, in which the number of stars indicates the rating on each factor.



Skincare brands	A. La Mer	B. Estee Lauder
Brand reputation	****	***
Research & development	****	***
Customer range	***	****
Marketing effort	***	****

Fig. 3. Task scenario.

To create gendered voices, we used Microsoft Azure text-to-speech service to create American English-speaking voices of a woman and a man. We then increased the pitch of a man’s voice, and decreased the pitch of a woman’s voice to create several gender-neutral voices for piloting. We finally picked the voice that had a mean gender perception closest to 50 (on a continuum of 0 (man) to 100 (woman)) with the smallest standard deviation ($M = 52.54$, $SD = 17.76$).

This series of pilot tests allowed us to create robust gender manipulations while keeping other aspects (such as perceived trustworthiness) consistent across agents of different genders. We acknowledge that there may be other ways to manipulate gender, such as using linguistic cues. For instance, women are more likely to use hedges to express ambiguity and indecisiveness [42, 44]. However, such linguistic manipulations of gender might risk rendering a woman agent as less

trustworthy than a man agent, thus introducing a confounding variable. Additionally, there is not enough evidence and research on the linguistic style of gender-neutral individuals.

3.2.2 Gendered topics and tasks. Based on prior research (e.g., [9, 31]), we identified two stereotypically feminine topics: skincare and cooking; two stereotypically masculine topics, sports and cars; and two topics that were found to have no strong gender associations: musical instruments and financing. We then created specific prompts about each topic using ChatGPT to generate realistic human-AI dialogues. The “gendered” topics and tasks can be seen in Table 1.

Gender Association	Topic	Specific Task
Gender-neutral (baseline)	Musical instruments	Finding a value-for-money piano model for beginners.
Gender-neutral	Financing	Finding a competitive small business loan option for your start-up.
Feminine	Skincare	Finding a skincare brand to negotiate a profitable endorsement deal for your social media influencer client.
Masculine	Sports	Finding a marketable American football star for your car company to sign an endorsement deal with.

Table 1. Gendered topics and tasks.

The resulting four topics and specific tasks were chosen based on a pilot test with 15 participants. As for perceived genderedness of the topics and tasks, the gender-neutral tasks were perceived as neither feminine nor masculine, with a relatively balanced ratio of men and women presumed to be familiar with the topic and task. The feminine task was perceived to be extremely feminine, with more women than men presumed to be familiar with the task; and the opposite for the masculine task. The responses generated by the agent and the table provided by the agent were perceived of equal quality across all four topics and tasks (see Appendix D Figure 27 for the statistics of repeated-measures ANOVAs and the sub-items for each dimension).

3.2.3 AI explanations. The XAI manipulation was intended to increase people’s awareness of how AI’s decision-making process is different from that of humans, to encourage analytical thinking and cognitive effort [15], and to prevent people from relying on heuristic thinking that may foster gender stereotyping the agent. To achieve these goals, for the XAI conditions, we adopted the feature contribution explanation [90] that quantifies the weight of each parameter in the final decision. Compared to other explanation styles and techniques that resemble how humans explain decisions (e.g., counterfactual explanation, case-based explanation), feature contribution explanation highlights the differences between AI and human decision-making. The AI explanations were first generated by ChatGPT and modified by the first author to incorporate feature contribution characteristics and to be task-specific. The feature contribution table and specific decisions were created by the authors. All of the explanation cards (see Figure. 4 and Appendix C) were pilot-tested for their quality along sub-dimensions of how much the explanation makes sense, the information trustworthiness, participants’ satisfaction with the explanation, and the perception that the agent knows what it is doing. No significant difference was found among all four task gender conditions (see Appendix D Figure 27 for statistics).

In addition to the feature contribution explanation specific to the decision, XAI manipulation was also done through adding generic explanations in the video vignettes explaining how the agent works. This was done by having the agent answer the question “How do you know all of these?” (see Appendix A).

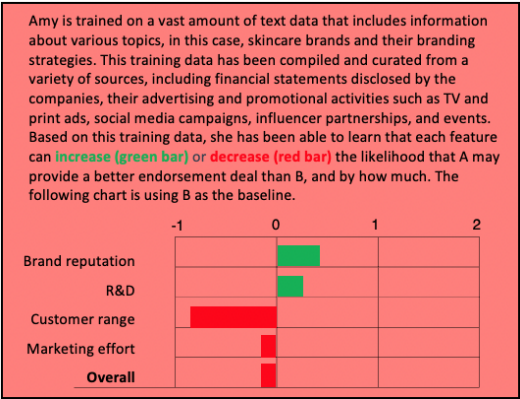


Fig. 4. The AI agent’s explanation.

3.3 Participants

Participants were recruited from Prolific, a large online crowdsourcing platform for data collection. Participants were compensated 10 dollars per hour for their time. A total of 412 participants based in the US participated in the study, among which 62 did not pass attention check or manipulation check. The remaining 350 participants included 171 identified as a woman, 169 men, 8 non-binary, and 2 prefer not to disclose gender. Their age ranged from 19 to 70 ($M = 40.57$, $SD = 14.44$). In terms of race/ethnicity, 247 (70.6%) identified as Caucasian/White, followed by 40 (11.4%) African American/Black, 27 (7.7%) Hispanic, 23 (6.6%) Asian, and 13 (3.7%) identified as “Other”. About half (50.9%) of the participants reported to be a Democrat, 57 (16.3%) Republican, 96 (27.4%) Independent, and 19 (5.5%) reported to have no political leaning. More than half (52.6%) of the participants reported to be not religious, 74 (21.1%) Protestants, 47 (13.4%) Roman Catholic, 4 Jewish, 4 Buddhist, 2 Muslim, 2 Mormon, 1 Orthodox, 1 Hindu, and 13 chose “Other”. Among the participants, 326 (93%) reported to have interacted with ChatGPT .

3.4 Procedure

Upon signing the informed consent, participants were presented with the first fictional scenario (Baseline Block) in which they needed to consult with our newly developed agent on buying a piano (a gender-neutral topic), and watched a video depicting their conversation with the agent (see Figure 5 Box B2; Appendix A). Then the narrative continued, in which the agent provided a table showing the ratings of each considered factor, for participants to compare two options (see Figure 5 Box B3; Appendix B). Participants were then asked to make their own decision, indicate their knowledge in this domain, and state their confidence in the decision. Then the agent’s decision was shown (with or without an explanation depending on the condition) (see Figure 5 Box B5; Appendix B). Participants were then asked to indicate their understanding of the agent’s decision-making process, mental effort in understanding it, whether they would like to change their decision to that of the agent, perceived humanlikeness, and gender perceptions of the agent.

The Baseline block with a gender-neutral task (the same task across all conditions) was intended to elicit pre-existing gender stereotypes toward the agent purely based on the gender and explanation manipulations of the agent, without priming them about which gendered topic the agent is familiar with.

Subsequently, participants were presented with a second scenario (Main Block) where the topic on which they consulted with the agent varied depending on the task gender condition. Before

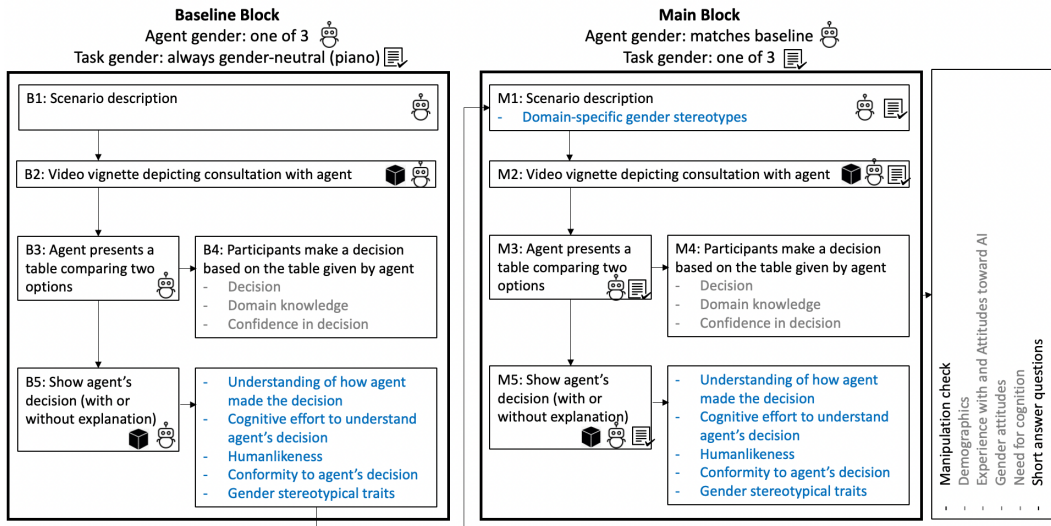


Fig. 5. **Procedures of the Online Experiment.** The robot icon indicates agent gender manipulation, the task icon indicates task gender manipulation, and the black-box icon indicates XAI manipulation. Bullet points are where measures occurred (blue bullet points measure dependent variables, and gray bullet points represent covariates).

watching the second video (see Figure 5 Box M2; Appendix A), they first answered questions that tested their domain-related gender stereotypes toward the agent (see Figure 5 Box M1). After that, they underwent the same procedure as they did for Baseline.

In the end, participants filled out manipulation check questions, demographics, AI attitudes, experience with AI, gender attitudes and Need for Cognition (NFC) scale. Then, they answered two short questions: 1) how did the gendered AI chatbot meet or violate their expectations and why; 2) whether they think any form of Artificial Intelligence (e.g., chatbots, robots, digital assistants) should be gendered and why? Lastly, they were asked to indicate their willingness to be contacted for a follow-up interview. Figure 5 illustrates the procedure.

3.5 Measures

3.5.1 Dependent variables.

Understanding of how the agent works. Participants self-reported level of understanding of how the agent works was measured using two items on a 7-point Likert scale: “Please indicate your level of understanding of how (agent name) processes information” (1= extremely low, 7=extremely high), “I understand how (agent name) came up with the prediction” (1=strongly disagree, 7=strongly agree). They were averaged to form a reliable scale (Cronbach’s Alpha= .92). In addition to subjective understanding, we verified participants’ objective understanding using one of the manipulation check questions that asked them to choose the correct statement about how the AI works from the following choices: “(agent name) has been trained on a large dataset of text from the internet”, “(agent name) thinks and reasons like a human”, and “No information was given regarding how it works”. Participants who chose the wrong answer were excluded from the data analysis.

Cognitive effort made to understand how the agent works. Adapting the NASA-TLX index [41], a single-item measure of mental demand was used to measure participants’ cognitive effort made to understand how the agent came up with the prediction. They responded to the question

“How much mental and perceptual activity was required (e.g., thinking, deciding, calculating, remembering, looking, searching, etc.) in understanding how (agent name) came to their decision?” on a scale of 1 (extremely low) to 7 (extremely high).

Perceived humanlikeness of the agent. This was measured using 3-item 7-point Likert scale. Participants were asked to indicate “how humanlike (agent name) is” (1=not humanlike at all, 7=extremely humanlike), “how (agent name) thinks like a human”, “how (agent name) thinks like you do” (1=not similar at all, 7=extremely similar). They were averaged to form a reliable scale (Cronbach’s Alpha = .89).

Domain-specific gender stereotype toward the agent. This was measured after participants finished the baseline block and before they watched the second video depicting the agent performing a task of a stereotypical gendered topic/domain. We measured participants domain-based trust in the agent as an indicator of gender stereotype. Depending on the topic gender condition, they were asked to rate “I think (agent name) would be informed about (topic)”, “I am willing to take (agent name)’s advice regarding (topic)”, “I DON’T think (agent name) would be knowledgeable about (topic)” (reverse-coded), “I think (agent name) is capable of finding me the information I need with regard to (topic)”, “I would trust (agent name) to help me (task)” (1=strongly disagree, 7=strongly agree). The 5 items were averaged to form a reliable scale (Cronbach’s Alpha = .91).

Perceived agent gender stereotypical traits. Participants were asked to rate the agent on 19 adjectives (traits) such as compassionate, aggressive, creative, etc. (1= extremely inaccurate, 7=extremely accurate). According to a recent meta-analysis of US public opinion polls [28], these traits formed three categories: communion traits (e.g., compassionate, affectionate, sensitive) were strongly associated with female stereotypes; agency traits (e.g., aggressive, dominant, ambitious) prevailed in male stereotypes; and competence traits (e.g., creative, intelligent, organized) used to be stereotypically male traits but are gaining gender equality over time and thus are considered gender neutral traits.

Conformity to agent decision. Participants whose decision was not the same as the agent were directed to answer the question of whether they would like to a) change their decision to what the agent suggested, or b) keep their own decision.

3.5.2 Covariates.

Domain knowledge. This was measured using a single-item 7-point Likert scale, “Please indicate your knowledge in this domain” (1= not knowledgeable at all, 7=extremely knowledgeable).

Need for cognition. We adapted the Need for Cognition Scale into a 6-item version [16]. Items included “I could prefer complex to simple problems”, “It’s enough for me that something gets the job done; I don’t care how or why it works” (reverse-coded). The six items were averaged to form a reliable scale (Cronbach’s Alpha = .82).

Attitudes toward AI. We adapted The General Attitudes towards Artificial Intelligence Scale (GAAIS) [80] into a 6-item version. Items included “I am impressed by what Artificial Intelligence can do.” “I think artificially intelligent systems make many errors” (reverse-coded). They were averaged to form a reliable scale (Cronbach’s Alpha = .76).

Attitudes toward sex-roles. We adapted Larson and Long [51]’s Attitudes toward Sex-Roles Questionnaire to measure participants’ pre-existing gender stereotypes. This included 8 items such as “Men make better leaders” (reverse-coded), “A man who has chosen to stay at home and be a house-husband is not less masculine.” (1=strongly disagree, 7=strongly agree). They were averaged to form a reliable scale (Cronbach’s Alpha = .83).

3.6 Semi-structured interviews and data analysis

To better understand and explain participants' responses in the online experiment, we also conducted 9 interviews with participants who completed the study and indicated interest in participating in the follow-up interview. Each of the 9 interviews lasted around 30 minutes. All the interviews were audio recorded and transcribed by the first author upon completion.

Our interview protocol included open-ended questions asking about participants' experience with any AI-powered device or technology, and their preference for the gender (i.e., gendered voice of Siri, gender of ChatGPT if they had options) of the device or technology. We probed for the underlying reasons for the preference, how the gender of the device or technology affects their use, perception and expectation of it, and whether and how their preference and expectation varies depending on what they use the device or technology for. Then, participants were asked how much they know about how AI systems work. After explaining and demonstrating (via an example shown in the study) what explainable AI is, participants were asked how the explanation affected their understanding, perception and expectation of AI-powered devices and technologies in general, and gendered AI technologies.

Following the qualitative thematic analysis guidelines [22], the first author conducted open coding on the interview transcripts and the short answers, and developed a loose array of open codes that were sorted through affinity diagramming. During this process, high-level themes and relationships between the themes emerged. Then, two of the authors iteratively examined the quotes in their context, and refined the codes and themes.

4 FINDINGS

4.1 Statistical Results

To choose the appropriate statistical method, we first conducted a series of Shapiro-Wilk tests to determine if the distribution of the variables was normal. Results suggested that for all the variables except for the 19 perceived agent gender stereotypical traits, a normal distribution can be assumed ($ps < .05$). For the 19 perceived agent gender stereotypical traits, the distribution departed significantly from normality ($0.72 < Ws < 0.88$, $ps < 0.01$).

We therefore performed non-parametric tests on the 19 gender traits, and parametric tests on all the other dependent variables. It should be noted that in this paper, we only report on the results of Main Block measures, because the baseline block was used to familiarize participants with the agent and to elicit domain-specific gender stereotypes, and thus did not have a task gender manipulation. However, the statistical patterns of Baseline measures resemble that of Main Block in understanding, cognitive effort, humanlikeness, and gender stereotypical traits.

4.1.1 Understanding of how the agent came to its decision. A 3 (Agent gender) by 3 (Task gender) by 2 (Explainability) Factorial ANOVA was performed on participants' understanding of how the agent works. Results suggested that only XAI had a significant main effect on the understanding ($F[1, 332]=50.32$, $p < .001$, $\eta_p^2 = .13$). Participants in the XAI conditions reported higher levels of understanding ($M=4.98$, $SE=.12$) than those in the non-XAI conditions ($M=3.81$, $SE=.12$). Agent gender ($F[2, 332] = .742$, $p = .48$) and task gender ($F[2, 332] = .168$, $p = .85$) had no significant main effect, and no significant interaction effect of any combinations of the independent variables ($ps > .21$, see Figure 6). Therefore, **H1** "Participants will have a higher level of understanding of how the AI agent works with (than without) explanations" is **supported**.

4.1.2 Cognitive effort in understanding how the agent came to its decision. Similarly, a 3 (Agent gender) by 3 (Task gender) by 2 (Explainability) Factorial ANOVA was performed on participants' cognitive effort. Results suggested that XAI only had a trend toward a significant main effect ($F[1,$

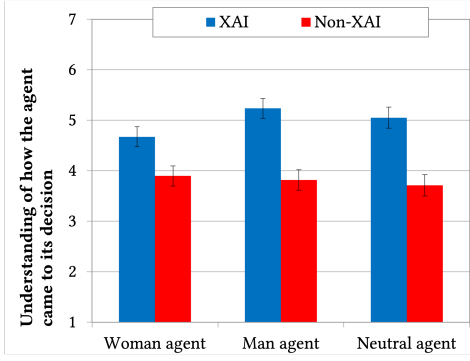


Fig. 6. Understanding of how the agent came to its decision, by XAI by agent gender (error bars represent standard error of the mean).

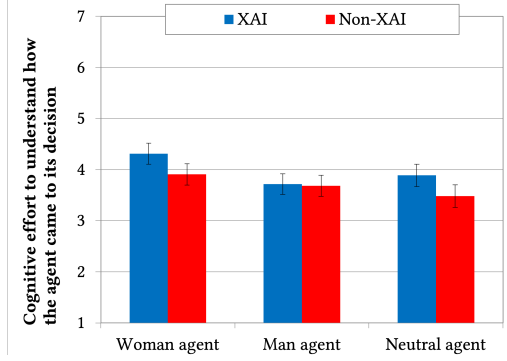


Fig. 7. Mental workload to understand how the agent came to its decision, by XAI by agent gender (error bars represent standard error of the mean).

332]=6.84, $p=.07$, $\eta_p^2=.02$, see Figure 7). Participants in the XAI conditions reported higher levels of mental workload ($M=3.96$, $SE=.12$) than those in the non-XAI conditions ($M=3.70$, $SE=.12$). Agent gender ($F[2, 332] = 2.23$, $p=.14$) and task gender ($F[2, 332] = .47$, $p=.63$) had no significant main effect, and no significant interaction effect of any combinations of the independent variables ($ps>.22$). **H2** “Participants will report greater mental workload in the XAI conditions than in the non-XAI conditions (H2)” is **supported** by weak evidence.

4.1.3 Perceived humanlikeness of the agent. A 3 (Agent gender) by 3 (Task gender) by 2 (Explainability) Factorial ANOVA was performed on perceived humanlikeness of the agent. Results suggested that only XAI had a significant main effect ($F[1, 332]=11.72$, $p<.001$, $\eta_p^2 = .034$, see Figure 8). Participants in the XAI conditions perceived the agent as less humanlike ($M=3.29$, $SE=.14$) than those in the non-XAI conditions ($M=3.98$, $SE=.14$). Agent gender ($F[2, 332] = 1.68$, $p = .19$) and task gender ($F[2, 332] = .30$, $p = .74$) had no significant main effect, and no significant interaction effect of any combinations of the independent variables ($ps>.24$). **H3** “Participants in the non-XAI conditions will perceive the agent as more humanlike than those in the XAI conditions” is **supported**.

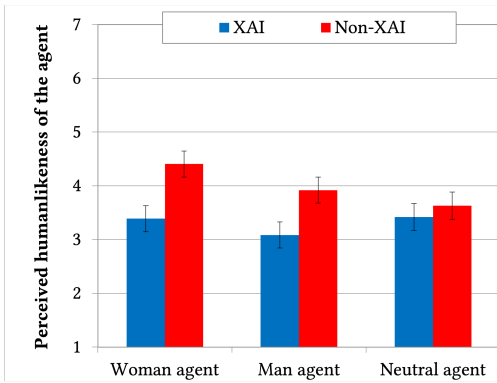


Fig. 8. Perceived humanlikeness of the agent, by XAI by agent gender (error bars represent standard error of the mean).

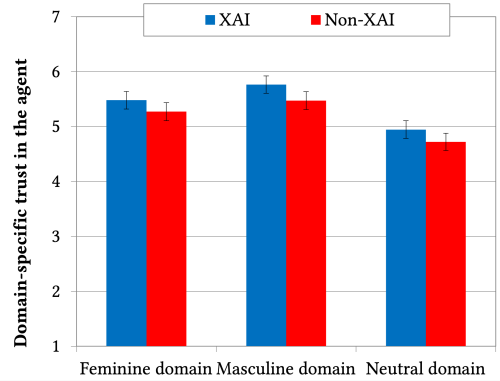


Fig. 9. Domain-specific trust in the agent, by XAI by task gender (error bars represent standard error of the mean).

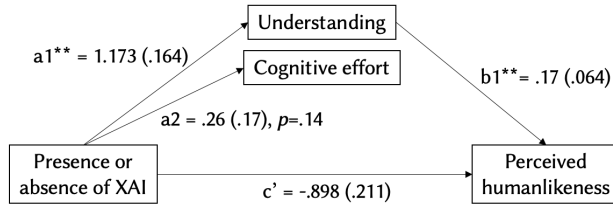


Fig. 10. Mediation analysis of the effect of XAI on perceived human likeness via understanding and mental workload.

4.1.4 Mediation of understanding and cognitive effort on the effect of XAI on perceived human likeness. To test the mediation effect, we followed Baron and Kenny [8]’s mediation analysis and interpretation guidelines. The results (see Figure 10) showed that there was a significant total effect between XAI and perceived human likeness of the agent ($B = -.699$, $SE = .199$, $p < .001$), and path a1 (i.e., XAI on understanding) ($B = 1.17$, $SE = .16$, $p < .001$) and path b1 (i.e., understanding on perceived human likeness) ($B = .17$, $SE = .064$, $p = .009$) were both significant. Finally, when understanding entered the relationship between XAI and perceived human likeness, the direct effect ($B = -.898$, $SE = .211$, $p < .001$) was significant. In addition, the Sobel test for the indirect effect is $z = 2.49$, $p = .013$; therefore, it was concluded that only partial mediation occurred between XAI on perceived human likeness via understanding. For cognitive effort, because path a2 (i.e., XAI on cognitive effort) ($B = .26$, $SE = .17$, $p = .14$) was insignificant, there was no ground for mediation. Therefore, **H4** “Participants’ understanding of how the gendered agent works and the cognitive effort will mediate the effect of XAI on their perceived human likeness of the agent” is **partially supported**.

4.1.5 Domain-specific trust toward the agent. A 3 (Agent gender) by 3 (Task gender) by 2 (Explainability) Factorial ANOVA was performed on domain-based trust in the agent. Results suggested that task gender had a significant main effect ($F [2, 332] = 19.05$, $p < .001$, $\eta_p^2 = .071$, see Figure 9). Post hoc comparisons using the Tukey HSD test suggested that participants trusted the agent significantly less in the gender-neutral task (i.e., finance) conditions ($M = 4.83$, $SE = .11$) than both the masculine task conditions ($M = 5.62$, $SE = .11$, $p < .001$) and the feminine task conditions ($M = 5.38$, $SE = .12$, $p < .001$). XAI had a trend toward a significant main effect ($F [1, 332] = 3.42$, $p = .06$, $\eta_p^2 = .01$). Participants trusted the agent more in the XAI conditions ($M = 5.41$, $SE = .09$) than in the non-XAI conditions ($M = 5.06$, $SE = .09$). Agent gender ($F [2, 332] = .87$, $p = .42$) had no significant main effect, and no significant interaction effect of any combinations of the independent variables ($ps > .42$). As a result, **H5** “the gendered agent will be trusted more in their respective gender stereotypical domain” was **not supported**, as there was no interaction effect between agent gender and task gender.

4.1.6 Conformity to agent’s decision. A logistic regression was performed to ascertain the effects of agent gender, task gender, the interaction between agent and task gender, AI explanations, participants’ domain knowledge, confidence in their decision, and need for cognition, on the likelihood that participants would conform to the AI’s decision or keep their own decision. When all the predictors were entered into the logistic regression model, it was statistically significant ($\chi^2 (12) = 45.705$, $p < .001$). The model explained 31.9% (Nagelkerke R Square) of the variance in participants’ conformity to AI’s decision and correctly classified 76.8% of cases. However, other than the interaction between agent and task gender, AI explanations, participants’ domain knowledge, the rest of the predictors were not significant ($ps > .15$).

We therefore performed a logistic regression with only the interaction between agent and task gender, AI explanations, and participants' domain knowledge entered into the model (see Figure 11), which was still statistically significant ($\chi^2(6) = 34.66, p < .001$). The model explained 25% (Nagelkerke R Square) of the variance in participants' conformity to AI's decision and correctly classified 74.4% of cases. Specifically, domain knowledge was significant in predicting the odds of an individual's keeping one's own decision ($b = .408, se[b] = .105, Wald \chi^2 = 15.046, p < .001$). For every one unit of increase in domain knowledge, there is a 50.4% increase in the likelihood of keeping one's own decision ($OR=1.504, 95\%CI [1.22, 1.85]$). As for the interaction between agent and task gender, compared to the baseline of a gender-neutral agent giving advice on a gender-neutral task, in the condition of a woman agent giving advice on a feminine topic, the odds of a participant to keep one's own decision is 0.229 times that in the baseline conditions ($OR=0.229, 95\%CI [0.07, 0.70]$). Explainability was also a significant predictor ($b = -.725, se[b] = .362, Wald \chi^2 = 3.908, p = .048$). In conditions where the agent provided explanations, the odds for participants to keep their own decision were 0.489 times that in the non-XAI conditions ($OR=0.489, 95\%CI [0.24, 0.99]$).

To examine the effect of XAI on gender stereotypes, we performed logistic regressions separately for XAI and non-XAI conditions, with predictors of the interaction between agent and task gender, as well as participants' domain knowledge (see Figure 11). The model for non-XAI was statistically significant ($\chi^2(6) = 34.434, p < .001$, Nagelkerke R Square = .493), but no longer significant for XAI ($\chi^2(6) = 8.423, p = .134$, Nagelkerke R Square = .117). Participants' domain knowledge remained a significant predictor in predicting their conformity to AI's decision for both XAI and non-XAI conditions ($ps < .011$). The interaction between agent and task gender was no longer a significant predictor ($Wald \chi^2 = 1.082, p = .897$) in predicting participants conformity to AI's decision for XAI conditions, but remained a significant predictor for that in non-XAI conditions ($Wald \chi^2 = 13.854, p = .008$).

Compared to the baseline of a gender-neutral agent giving advice on a gender-neutral task, in the condition of a woman agent giving advice on a feminine topic, the odds of a participant to keep one's own decision is 0.05 times that in the baseline conditions ($OR=0.05, 95\%CI [0.007, 0.346], p = .002$), and in the condition of a man agent giving advice on a masculine topic, the odds of a participant to keep one's own decision was 0.162 times that in the baseline ($OR=0.162, 95\%CI [0.036, 0.715], p = .016$). In other words, only in non-XAI conditions, were participants more likely to conform to the agent's decision when the gender of the agent matched that of the task. The statistical results provide evidence for **supporting H6** "Participants in the non-XAI conditions will more likely conform to the gendered agent's decision in their respective gender stereotypical domain than those in the XAI conditions".

4.1.7 Gender stereotypical traits. Because normal distribution for all the gender stereotypical traits could not be assumed, we performed a series of non-parametric tests, specifically Kruskal-Wallis tests that allowed for comparing the difference across 3 agent genders. Results suggested that, for agents that provided explanations (XAI conditions), no significant difference was found across 3 agent genders on all 18 traits ($Hs(2, 176) > 4.37, ps > .11$, except for "decisive", $H(2, 176) = 7.55, p = .02$ (see Figure 12). Post hoc comparisons using Mann-Whitney Tests (with a Bonferroni correction adjusted alpha level of .017 (.05/3) suggested that explainable gender-neutral agent was perceived as significantly more decisive (mean rank=104.24) than an explainable woman agent (mean rank=76.81, $z = -2.74, p = .018$). None of the other comparisons were significant after the Bonferroni adjustment ($ps > .36$).

On the other hand, for agents that did not provide explanations (non-XAI conditions), there were significant differences across agent genders on almost all the stereotypically feminine traits such as "affectionate" ($H(2, 176) = 7.09, p = .03$), "compassionate" ($H(2, 176) = 7.47, p = .02$), "sensitive"

XAI								
Predictors	B	S.E.	Wald χ^2	df	Sig.	Exp(B)	95% C.I. for EXP(B)	
							Lower	Upper
interaction between agentgender * taskgender								
(neutralgender*neutraltask as baseline)			1.082	4	0.897			
[woman agent] by [feminine task]	-0.163	0.781	0.043	1	0.835	0.85	0.184	3.925
[woman agent] by [masculine task]	0.895	0.923	0.941	1	0.332	2.448	0.401	14.947
[man agent] by [feminine task]	-0.032	0.647	0.002	1	0.961	0.969	0.272	3.446
[man agent] by [masculine task]	-0.055	0.688	0.006	1	0.936	0.946	0.246	3.644
domain knowledge	0.307	0.121	6.489	1	0.011	1.36	1.073	1.722
Constant	-0.916	0.464	3.896	1	0.048	0.4		
Test			Wald χ^2	df	Sig.			
Omnibus tests of model coefficients			8.423	5	0.134			
non-XAI								
Predictors	B	S.E.	Wald χ^2	df	Sig.	Exp(B)	95% C.I. for EXP(B)	
							Lower	Upper
interaction between agentgender * taskgender								
(neutralgender*neutraltask as baseline)			13.854	4	0.008			
[woman agent] by [feminine task]	-2.992	0.984	9.236	1	0.002	0.05	0.007	0.346
[woman agent] by [masculine task]	0.324	1.253	0.067	1	0.796	1.382	0.119	16.122
[man agent] by [feminine task]	0.048	1.221	0.002	1	0.969	1.049	0.096	11.485
[man agent] by [masculine task]	-1.823	0.759	5.766	1	0.016	0.162	0.036	0.715
domain knowledge	0.638	0.224	8.127	1	0.004	1.893	1.221	2.934
Constant	-0.364	0.614	0.351	1	0.554	0.695		
Test			Wald χ^2	df	Sig.			
Omnibus tests of model coefficients			34.434	5	<.001			

Fig. 11. Logistic regression analysis of participants' conformity to agent's decision, by XAI conditions.

N=161 (only those whose decision was different from the agent's)

		df	XAI		Non-XAI	
			Kruskal-Wallis H	Asymp. Sig.	Kruskal-Wallis H	Asymp. Sig.
Stereotypically feminine traits	affectionate	2	0.417	0.812	7.092	0.029
	compassionate	2	1.185	0.553	7.468	0.024
	sensitive	2	0.535	0.765	9.925	0.007
	emotional	2	0.389	0.823	7.567	0.023
	understanding	2	1.423	0.491	3.507	0.173
	tender	2	0.933	0.627	6.034	0.049
	sympathetic	2	0.095	0.953	7.879	0.019
Stereotypically masculine traits	ambitious	2	0.233	0.89	4.351	0.114
	aggressive	2	1.011	0.603	6.263	0.044
	courageous	2	0.446	0.8	6.702	0.035
	decisive	2	7.546	0.023	2.183	0.336
	dominant	2	2.209	0.331	1.537	0.464
Gender-neutral traits	competitive	2	1.944	0.378	3.109	0.211
	creative	2	0.788	0.674	5.016	0.091
	intelligent	2	2.582	0.275	3.75	0.153
	innovative	2	2.091	0.352	1.579	0.454
	organized	2	4.374	0.112	1.288	0.525
	competent	2	2.323	0.313	4.131	0.127
	informed	2	2.398	0.254	4.96	0.102

Fig. 12. Kruskal-Wallis tests results of perceived gender stereotypical traits by agent gender by XAI conditions.

XAI							
Avatar gender	Mean Rank			Test	Std. Test	Sig.	Adj.
	woman	man	neutral	Statistic	Statistic		Sig.
Decisive	76.81	87.79	104.24	-25.425	-2.744	0.006	0.018**
Non-XAI							
Avatar gender	Mean Rank			Test	Std. Test	Sig.	Adj.
	woman	man	neutral	Statistic	Statistic		Sig.
Affectionate	97.17	90.33	73.61	23.556	2.603	0.009	0.028*
Compassionate	97.08	90.89	73.09	23.982	2.649	0.008	0.024*
Sensitive	98.48	91.43	70.93	27.557	3.052	0.002	0.007**
Emotional	97.64	89.95	73.51	24.132	2.706	0.007	0.02*
Tender	95.68	91.1	74.41	21.276	2.35	0.019	0.056
Sympathetic	96.25	92.37	72.37	23.88	2.636	0.008	0.025*
Aggressive	80.58	99.25	82.14	-18.675	-2.27	0.023	0.07
Courageous	79.22	99.93	82.89	-20.717	-2.436	0.015	0.045*

Fig. 13. Pairwise comparisons of gender stereotypical traits of agents, showing test statistics for highlighted pairs that yielded significant difference.

($H(2,176)=9.93$, $p=.01$), “emotional” ($H(2,176)=7.57$, $p=.02$), “tender” ($H(2,176)=6.03$, $p=.05$), and “sympathetic” ($H(2,176)=7.88$, $p=.02$). Post hoc comparisons using Mann-Whitney Tests (with a Bonferroni correction adjusted alpha level of .017 (.05/3) suggested that for all of these stereotypically feminine traits, the woman agent was rated significantly higher on these traits than the gender-neutral agent. For stereotypically masculine traits “aggressive” ($H(2,176)=6.26$, $p=.04$) and “courageous” ($H(2,176)=6.70$, $p=.04$), the man agent was rated significantly higher than the woman agent (see Figure 13 for pairwise comparisons). These results **support H7** “The gendered agent will be more likely to be assigned their respective gender stereotypical traits in the non-XAI conditions than in the XAI condition”.

4.2 Qualitative Results

The statistical results suggested an inconsistency in participants’ gender stereotypes toward the agent. Specifically, while participants did not report higher trust in the gendered agent in the respective gender stereotypical domain, gender stereotypes were evinced more subtly such that they did tend to conform to the AI’s decision in gender-matching conditions and tend not to conform in gender-swapped conditions. In part to reconcile such inconsistency and to better answer RQ0 (*how do people apply gender stereotypes toward gendered AI*), we present our qualitative findings around conscious awareness of explicit gender stereotypes and potential unawareness of implicit gender stereotypes toward AI agents. Additionally, we also present findings on participants’ perceptions of how XAI could potentially mitigate gender stereotypes toward AI agents to better answer RQ1.

We annotate the source of the quotes by indicating their gender (W=woman, M=man, N=binary or third gender, or “Prefer not to disclose”), age, and experiment condition (Agent gender: 1=woman, 2=man, 3=neutral; Task gender: 1=feminine, 2=masculine, 3=neutral; XAI: 1=XAI, 2=non-XAI), followed by the interviewee number (Px) if the source is the interview.

4.2.1 A clear awareness of and opposition against gender stereotypes toward gendered devices.

Our qualitative data shows that some people are well aware that matching the agent’s gender with the task it is designed to perform can reproduce and reinforce existing gender stereotypes. For example, a participant expressed their strong objection to the question of whether any form of AI should be gendered and why,

No, misogyny (Prefer not to disclose, 26, 222)

Many others shared the same belief that the dominating trend and default of gendering digital assistants as female has contributed to and will continue to propagate gender stereotypes and gendered expectations in society.

Gendering them female probably only contributes to the subjugation and servitude of women. (W, 37, 132)

The tendency for bots to be female upholds stereotypes of women being submissive and beholden to authority. (N, 22, 211)

I do not think AI should be gendered because it will keep perpetuating antiquated gender norms into the digital environment. For example, Siri typically has a female sounding voice and is used for more traditionally feminine tasks like scheduling on the calendar or making reminders or shopping lists. (W, 32, 132)

Some also noted that gendering an agent might introduce gender-associating bias regarding the information provided. While we could not probe for further elaboration for these short answers, the following response clearly suggests an awareness of potential gender stereotypes toward AI agents.

I do not think they should be gendered. Though this does make Artificial Intelligence seem more human, it allows for unnecessary prejudice and stereotyping to play a role in what answers are being provided and by “who.” (N, 25, 212)

Being aware of the gender stereotypes imposed by having AI agents perform its gender stereotypical task, participants expressed aversion to this practice.

I would think that doing gender by tasks would take us more into, all the housekeeping tasks would be female voices, and we’re trying to steer clear of that sort of gender roles. (W, 56, 132, P7)

Some reported that they would like to play with the gender of the tool to rebel against the traditional gender norms and expectations by having the AI agent do tasks that are counter-stereotypical. The following participant elaborated on why she would prefer a female bot that provides medical advice,

With regards to ChatGPT and for any kind of AI, that’s why I like to mix it up. I don’t want to fall into that rut of thinking that it’s only men or it’s only women for these certain tasks. I might choose a woman (for a doctor bot) just because I’d want to get away from the patriarchy impression that all doctors are men [...] I was deliberately choosing something that’s less commonly viewed as a woman’s role, because that’s the kind of thing I’d want to rebel against. And I don’t want to think that all authority voices are male. (W, 62, 321, P2)

Interestingly, all the responses to the short answer questions that demonstrate participants’ awareness of gender stereotypes toward gendered AI are from women or non-binary individuals or participants who preferred not to disclose gender, despite our best effort to dig for men’s responses that shared that perspective. That said, this is not to be taken as men are more likely to possess gender stereotypes toward AI agents than women do, or as most men are unaware of their gender stereotypes toward AI agents. After all, the quantitative results did not show evidence for such a gender difference. Instead, in the interviews, some men tried to express their gender unbiased attitude as well, even though this could be due to the interviewer’s gender being female. One male participant expressed that humans and technologies alike, should not be judged by their gender.

The device is designed to do a job, and it doesn’t matter if it’s male or female. That should be true for humans, too, but it obviously isn’t completely true these days. It’s getting better, I think. (M, 70, 131, P4)

4.2.2 *Implicit gender stereotypes toward gendered technologies.*

While many participants indicated their conscious awareness and concern of extending gender stereotypes to gendered technologies, it does not mean that being consciously aware of the gender stereotypes naturally makes the person free of implicit gender stereotypes toward gendered technologies. For instance, while P6 said earlier in the interview that he was opposed to the idea of traditional gender roles being preserved in AI, when asked about his preference for the gender of the AI across different modalities, his response implied a touch of misogyny.

I don't know if I was having surgery, I might feel more comfortable having a male talk to me [...] I think a male robot would be a little more comfortable for some people for surgeries [...] If there's a gendered robot, I would hope that they're a lot more advanced. I wouldn't want to have the same level of technicality as a chatbot. I think a chatbot would be way more dumb down, so to speak [...] the severity of what the robot has to do, because surgery is a lot more important and vital than customer service [...] I'd rather have a female customer service than a male, and then for surgery, I'd rather have a male. But that's just personal preference. (M, 30, 112, P6)

Even though P6 tried to explain away his implicit gender bias by emphasizing that it was just his personal preference, the interviewer, by continuously probing for the underlying reasons for his preference, was able to piece out a rather misogynistic account in which female was associated by him with dumb-down customer service and male associated with "more advanced" surgeons.

The responses to the short answer questions are also filled with examples of implicit gender stereotypes. For instance, contrary to some participants' (mostly women or non-binary individuals) awareness of the social consequence of digital devices being too frequently feminized, many men expressed content with and ready acceptance of the practice. For instance, in responding to whether the agent met their expectations, several participants explicitly pointed out that the female gender of the agent met their expectations.

She is a girl so meets my expectation. (M, 38, 111)

Some participants also indicated that AI should be gendered, reasoning that certain gender can help more with certain tasks.

Yes, there are different applications to where it may be used to help people and specific genders can help more. (M, 27, 222)

Some elaborated on what gender might help more with what gender stereotypical tasks, stating that this was not necessarily his own opinion, but what he thought others would feel comfortable with.

It really doesn't have to (be gendered), but I can envision others would prefer them in gendered roles based on what the subject is—a man for things like sports, repairs and automotive, and a woman for things like fashion and beauty/cosmetics. This might make the person using it more comfortable overall. (M, 50, 221)

One young man explained how being consciously aware of one's gender associations is one thing, but trying to not let the entrenched associations affect his perception and behavior is another. He started by reflecting on the effect the gender of the voice might have on his perception of the information. He admitted that despite feeling the need to be open-minded, he still could not control his "innate" tendencies to evaluate information based on the gender of the source.

If I'm getting advice on more feminine aspects, whether it be related to fashion or cooking or child care, I would feel like there's insight there that a masculine voice wouldn't have... but is it fair to associate a feminine voice with questions about child care? It's like that kind of sexism is so deeply rooted to human psyche [...] It does feel

like I am innately driven to ask a guy for things But at the same time, I feel like I should be more open minded But I am more likely to listen to a masculine voice when it talks about certain things, I'm more likely to absorb it better. And that's not something I really have control over. (M, 33, 232, P5)

He further reasoned that this was because it is the way the gendered reality is: from how he was raised, to the media contents he consumed, traditional gender associations are infiltrated in every aspect of life. The gendered reality has shaped and reinforced his perceptions and perspective, a gendered reality that he "can't escape".

I find myself drawn to certain voices talk about certain subjects. For example, if I'm listening to a video game like a review or something, it almost always is a guy. If I'm asking advice on how do I cook a gluten-free dairy-free cake, whenever I Google that I'm brought to all these different blogs written by women. So I would expect to have it be read in their voice. If I'm reading something about child care which I don't personally know much about, I would imagine you would have a mother's voice to it, and it wouldn't just be any girl's voice. It'd be a mother's voice... Whether it be stereotypical or not, if I ask my mom about childcare, she would know everything about that. It's just the way I was raised, I suppose it was very normal. I guess you can't escape that. (M, 33, 232, P5)

This gendered reality extends to the perception of modern technologies, such that the conscious awareness that AI will provide the same information regardless of its gender is often defeated by the subconscious feeling that gender does play an important role in information judgment.

But since we're talking about AI, it's like I know that the AI is going to have the exact same information available to it. It's like my heart is saying, Yes, it (gender) kind of does matter. But my logic, my brain is saying No, of course not. (M, 33, 232, P5)

As for the gender stereotypes toward the non-binary gender agent, quite a few participants indicated their conservative feelings about having an agent of the third gender.

As per my belief, I actually don't believe in the third gender. (M, 35, 311)

If they are to be gendered in any way I think it should be limited to just two. (M, 51, 122)

I think it should be just 2 genders, male and female. (M, 38, 121)

One alluded to her stereotype about trans-gender individuals to be offensive.

He was introduced as trans-gender, and his look and voice created that effect, but I was surprised that he was not offensive. (W, 50, 311)

4.2.3 Perceived potential of XAI to mitigate gender stereotypes toward AI agents.

In the interviews, participants were shown an example of the explainable AI that half of them had seen in the online experiment. Many commented on how the AI's explanations rendered the gender of the AI irrelevant. The explanations can draw participants' attention away from the gender of the AI, and toward the information and data that seems impartial with the explanation. AI's decision is therefore perceived as more trustworthy because it accounts for and is based on the synthesis of impartial data. For instance, P6 reported that gender would not matter as much with XAI,

It (XAI) makes a lot of sense. From how much data it can look at. It actually makes me a little bit more likely to trust an AI decision more than a human based on all the input it can receive. So I think it makes it more trustworthy. And I don't think gender should matter as much knowing that you're gonna get a pretty smart opinion based on all the input. So the gender really isn't as relevant... Regardless of the gender, you're going to

get a less biased opinion. You're gonna get a really well informed opinion based on all the factors that it plays into for what it looks into. (M, 30, 112, P6)

Others shared the same view that explanations can increase people's awareness that the AI agent, regardless of its anthropomorphic gender representation, is a piece of technology after all, that functions differently from humans. Even if some might believe that a certain gendered agent is better at certain gender stereotypical tasks, seeing how the information is presented and factored into the decision could eliminate such thought. As the following quotes illustrate,

Personally if I knew that the bot was programmed to do that correctly, I would trust either gender. There are some people that have that wired into them that females can't do mechanics, or men can't do child care or cooking, so they would trust the stereotypical gender to (give advice) more than not. If it's a program to function as technology, then I don't think gender matters. (W, 57, 332, P8)

I think the explainable AI, where you're letting everyone know how the decisions are coming to, what factors are being put in the decisions. The chance that you would feel like you needed a gender to be applied in any scenario is, I would say that mitigates the need for gender. If you have all the information like it's presented here about how these decisions are made, you're not going to think of it as a gender. You're going to think of it as data as facts, just like a calculator, and you just put in the numbers you need to fix and it comes out. The calculator isn't going to give a different answer, no matter who uses it or who made it. It's just a machine. (W, 56, 132, P7)

For P5 who found it hard to overcome his "innate" tendency to be drawn toward listening to and better absorbing certain information coming from a certain gender, he indicated that he saw explainable AI as the AI's credentials on the wall that could convince him of its ability and knowledge in areas he would not otherwise entrust it to provide advice.

For example, a masculine voice talking about something that has a feminine aspect to it, and it explains it. I think it would help people. When it comes to childcare, most people aren't going to ask the guy how to do that. But if the guy would have the credentials, if he were like a doctor who specializes at those, people like, oh, okay, I'm going to listen to. So this is basically the AI's credentials. It's saying "I know what I'm talking about." That's how I see it. It is their credentials on the door. It's their big Harvard on the wall, the diploma on the wall. (M, 33, 232, P5)

While most participants expressed optimism about the potential of XAI to mitigate gender stereotypes toward gendered AI agents, some cautions were raised realizing that not all people are alike and that deeply entrenched sexist beliefs may not be easily uprooted. P8 noted that whether XAI can mitigate or reinforce gender stereotypes will depend on whether the person is open-minded enough to change their way of thinking.

It has the potential to go either way, to increase those gender stereotypes and to mitigate them. Some people will look at it and be like well see that's totally female, that's totally wrong, that's not how I would do it. Of course the females doing it so it's wrong. And there goes, the stereotype continues. Some people will be like, now I see it's just information. It's not necessarily male or female. And then you're mitigating the gender stereotypes. Just we're in such a divisive world right now. It's more that people have an open mind to accept new information and change the way they think, or they have a closed mind and don't want to accept new information or change the way they think. (W, 57, 332, P8)

P2 brought up an interesting perspective and caution that, because women are stereotypically more likely to explain themselves, XAI may be perceived by some to be possessing such “feminine” qualities, and thus be assigned the female gender.

I wonder if there’s a persistent belief that by definition XAI will seem biased to some, specifically the belief that XAI is female in gender because of the belief that women explain their decisions more. (W, 62, 321, P2)

5 DISCUSSION

In this study, we sought to examine whether and how people apply gender stereotypes toward gendered AI agents that are representative of large language models similar to ChatGPT (RQ0) and explore the effect of eXplainable AI on mitigating those gender stereotypes (RQ1). Findings suggest that participants indeed applied gender stereotypes toward gendered agents, in that they were more likely to conform to the AI’s decision regarding a gender-matching topic (i.e., woman agent giving advice on a stereotypically feminine topic), and that they rated the woman agent higher on stereotypically feminine traits (e.g., emotional, sensitive), and the man agent higher on stereotypically masculine traits (e.g., aggressive, courageous). Further, XAI significantly increased participants’ understanding of how the gendered AI agent came to its decision and decreased their rating of the agent’s humanlikeness, thereby mitigating their gender stereotypes. We return to our research questions to discuss and interpret what these findings mean in light of existing literature. In doing this, we also suggest directions for future work to explore.

5.1 Subconscious automatic application of gender stereotypes toward AI agents

In response to RQ0 (Do people apply gender stereotypes toward AI agents?), both the statistical evidence and qualitative data suggest that individuals apply gender stereotypes toward agents **without realizing that they hold these stereotypes**. This is evident in the findings that only two (conformity and gender stereotypical trait ratings) of the three gender stereotype indicators yielded significant results while domain-specific trust did not show any indication of participants applying gender stereotypes. This seems contradictory because conformity to the agent’s decision should logically follow trust in it. One explanation pertains to the way domain-specific trust in the gendered agent was operationalized. By explicitly linking the gendered agent to a stereotypically gendered task (i.e., Amy would be informed about skincare brands), the questions may have flagged to participants as an assessment of their gender stereotypes and thus may have led to socially desirable answers [30] or caused psychological reactance [81]. As a result, participants might have approached these questions more mindfully and corrected the inappropriate gender associations of which they came to be aware due to the salience of gender cues. The conformity and gender stereotypical trait questions, however, did not make the gender cues salient. When answering these questions, participants may not have come to realize their dormant gender stereotypes. Such unawareness of the activation of implicit gender stereotypes can often account for the application of them [34]. Lee [55]’s work also suggests that the very awareness of one’s gender stereotypes could suppress one’s tendency to apply and display those stereotypes. Our qualitative findings corroborate this explanation in that when explicitly asked about their gender associations with tasks, participants straightforwardly denied they had any. But their responses to other subtle questions gave away their implicit gender bias (see Section 4.2.2 for examples).

Essentially, findings of this study are in line with the large body of prior work demonstrating that individuals do apply gender stereotypes to AI agents (e.g., [21, 68]), and counter sporadic exceptions [14, 65]. However, this study provides nuanced methodological insights that can serve to reconcile the inconsistency in literature. Drawing on our own seemingly contradictory findings

obtained through the variety of methods to operationalize gender stereotypes, we suspect that it was because the gender stereotype operationalizations revealed the assessment purpose that the outlier studies failed to observe gender stereotypes toward virtual agents or robots. For instance, Bryant et al. [14] employed a measure similar to our domain-specific trust in gendered robots that suggested participants applied no gender stereotypes; and Nag and Yalçın [65] employed a within-subjects study design that could have led to sensitization effect [35] such that a gradual awareness of the study's objectives may have altered participants' responses. This suggests the benefit for future studies investigating the stereotyping of computer agents to employ stereotype operationalizations that are less likely to bring stereotype to conscious awareness, such as using a between-subjects design, or more discreet measures of stereotypes (e.g., Implicit Association Test [36]).

To re-emphasize, the gender-neutral agent in our paper was portrayed as a gender non-binary agent, toward which participants also held gender stereotypes (e.g., perceiving it to be less compassionate and more aggressive than a woman-looking agent). This has demonstrated that gender-neutralizing AI agents does not eliminate but rather evokes gender stereotypes. However, gender-neutralizing is different from de-gendering AI agents, wherein the agent is presented as lacking any gender characteristics, or genderless, as seen in some LLMs (e.g., ChatGPT, Bard). While our study did not provide direct evidence for supporting either that genderless agents could prompt or they could prevent gender stereotypes, both our qualitative findings (e.g., P5) and recent literature [61, 84, 86] have suggested that individuals' association of a certain topic or activity with a particular gender can be triggered in the absence of explicit gender cues. Further, the agent's use of natural language increases individuals' likelihood to apply heuristics and social characteristics to it [55, 68] despite its being genderless. It is conceivable that people perceive genders in genderless AI agents. However, to empirically test this, future work should examine the perception of agent gender and stereotypes in genderless agents by stripping all explicit gender cues.

5.2 Leveraging XAI to correct social harms caused by gendered AI technologies

While there has been a growing interest in providing AI explanations that match human cognitive and reasoning patterns [59, 60, 62, 89], our work demonstrates the value of XAI techniques that purposefully diverge from human cognitive style. Specifically, the XAI technique employed in this study (i.e., feature contribution) effectively mitigated individuals' gender stereotypes toward AI agents through improving their understanding of how the agent came to its decision (RQ1).

Participants' understanding of how the agent works partially mediates the effect of XAI on perceived humanlikeness of the agent, and their subsequent gender stereotypes toward it (H4). This implies that participants likely engaged in analytical thinking that prompted them to mindfully acknowledge the agent's distinction from humans, which in turn inhibited the use of representativeness heuristics due to a salient lack of similarity [87] between humans and agents. This mitigation pathway via analytical thinking and inhibition of heuristics substantiates the efficacy of recent efforts [89] in proposing XAI techniques to alleviate heuristic biases. Unlike the techniques proposed in [89] that deliberately mimic human generated explanations, we elected to utilize the feature contribution XAI technique to purposefully diverge from human thinking, as the heuristic bias we target is toward the AI itself. As such, we emphasize the importance of designing XAI techniques that account for social and contextual differences instead of applying universal principles that disregard specific contexts.

Briefly returning to the motivation that drove this inquiry – leveraging XAI to mitigate the cognitive bias underlying both a) gender stereotyping and b) individuals' application of social characteristics to AI agents. Admittedly, our XAI intervention only undermines and disrupts the mechanism that accounts for the latter. By highlighting the distinction between humans and agents,

it obstructs the transfer of gender stereotypes from humans to agents. Even though this does not directly tackle the underlying source of gender stereotypes toward humans, it minimizes the reinforcement of existing gender stereotypes and expectations, which occurs through prolonged interactions with anthropomorphic gendered AI agents that often uphold traditional gender norms [17, 85]. Our approach serves as an intermediary solution to correct and remedy the social harms caused by the technologies designed without careful consideration of social equality, justice and diversity [25]. As an instantiation of responsible XAI [6], its application in mitigating gender stereotypes also echoes the appeals for design principles that reflect feminist values and convictions [7, 88] that HCI community has sought to leverage to bring about social change and create more equitable, just, and inclusive sociotechnical future.

5.3 Design implications

Results from this study directly inform how to design AI-powered technologies to minimize users' application of gender stereotypes toward them. We first suggest several ways this can be achieved. However, mitigating gender stereotypes toward AI agents represents merely a small step in the pursuit of a comprehensive approach to tackling the broader challenge of gender stereotypes directed at humans. Therefore, we also discuss implications of our findings for the design of technologies that possess the potential for a wider social impact in mitigating gender stereotypes toward humans.

5.3.1 Explaining away gender bias. Our findings demonstrate that the feature contribution XAI technique has rendered the gender of the AI agent irrelevant. To mitigate gender stereotypes toward them, future AI agents derived from large language models should consider implementing such an explanation method, which could be done through following up the agent's response to users' prompt with a decomposition of the parameters played into the response, or the sources it extracted information from to synthesize the response and its evaluation of the reliability of the sources, as well as the weight each source of information contributed to the response. In other words, in addition to the response in natural language (which makes it humanlike and thus likely to evoke gender stereotypes), provide information that emphasizes its distinctive non-humanlike attributes, such as the ability to rapidly process large quantifiable data. Alternatively, such an intervention can occur even before gender stereotypes begin to form toward gendered chatbots, which can be done through transparent interface design [57]. Take chatbot interface as an example, before generating responses to the user's prompt, the interface could visually display and animate the bot's information retrieval, source evaluation, and parameter definition process, along the lines of infographics design [40]; or verbally describe such processes by simply providing peripheral responses like "accessing X sources", "synthesizing information from sources exceeding the reliability threshold of X", "generating X parameters".

5.3.2 Purposefully gendering AI agents to counter gender stereotypes. Finally, while it is heuristic thinking that accounts for stereotyping and social perceptions of AI agents, such heuristics can be nevertheless leveraged to challenge gender stereotypes by rewiring gender associations through progressive design of AI technology. As our qualitative data showed, individuals' gender associations are largely shaped by their experience and the environment, including their upbringing, daily encounters performing traditional gender roles and occupations, and their daily media consumption that confirms their gender stereotypes (i.e., woman bloggers who write about recipes, man are game commentators). While technology cannot change one's upbringing, it is possible to create a progressive gendered reality that can reshape human experience that will hopefully correct the gender stereotypes. For instance, with the advancements in large language models that users consult on a wide range of topics, it will be beneficial to allow the detection and categorization of

the “genderedness” of the user’s prompts, and gender the responding agent in a gender-swapped manner. As an example, when a user consults the AI on a stereotypically feminine topic (e.g., cosmetics, child care), the system will assign a man agent to respond rather quickly with in-depth and trustworthy answers to demonstrate “his” expertise in that domain. This gender-swapping can manifest in the agent’s linguistic style as well. For instance, a man agent could use more politeness markers, hedges, paralinguistic cues that are more often used by women [42, 44]; and a woman agent could sound more assertive, and use fewer qualifiers and disclaimers [29, 46], to reverse gender stereotypes. Additionally, for certain users displaying tendencies of discrimination or bias toward gender non-binary individuals, perhaps manifested through toxic comments in social networking sites (although this approach raises privacy concerns necessitating future work to balance the trade-off), a gender-fluid agent may be used to correct such bias, as increasing the frequency of contact with stereotyped individuals could potentially reduce stereotypes [4].

6 LIMITATIONS AND FUTURE WORK

The findings of this study need to be interpreted with several limitations in mind. These include the form of the AI, and the specific AI explanation technique. First, while we were able to ascertain that XAI had an impact on reducing participants’ gender stereotypes toward a chatbot similar to that of ChatGPT, the same effect may not be obtained for other forms of AI. For instance, for mechanical robots that have a heavy bold physical embodiment, individuals’ association of them with masculinity [82] may not be easily explained away by XAI. This suggests an opportunity for future work to look into ways to reduce gender stereotypes toward physically embodied robots. Second, we used a specific type of XAI technique – feature contribution – as a means to differentiate AI decision-making process from that of humans, in order to disrupt the application of representativeness heuristics. It is worthwhile for future work to investigate whether other types of XAI techniques, such as those resembling humans’ explanation of decisions (e.g., counterfactual case, examples) have the same effect. Or due to their resemblance to humans’ thinking process, these techniques might make humans more likely to apply representativeness heuristics, thereby exacerbating gender stereotypes toward the AI. Lastly, to balance the trade-offs between the level of manipulation control and realism, we chose to use pre-recorded video vignettes to elicit relatively genuine responses from participants. However, such an approach may not have adequately and immersively engaged participants in the depicted scenario to evoke the wide range of authentic and contextual reactions they may have in more interactive live settings. This limitation prompts thought-provoking inquiries about whether, in a more interactive setting, participants will be more or less likely to apply gender stereotypes toward a gendered AI agent? And will feature contribution XAI technique be more or less effective in discouraging people from applying these stereotypes? These questions pose avenues for future work to investigate.

7 CONCLUSION

Gender stereotypes are carried over to humans’ interactions with artificial intelligence. In this study, we examined the effect of XAI on the mitigation of gender stereotypes toward gendered AI agents. We conducted an online experiment using video vignettes, in which we manipulated the gender of the agent, the stereotypical gender of the task the agent helped with, and the presence or absence of the AI agent’s explanation using feature contribution XAI technique. We found that XAI significantly increased participants’ understanding of how the agent came to its decision and decreased the perceived humanlikeness of the agent. Only in the absence of AI explanations did we observe participants’ gender stereotypes toward the agent. These gender stereotypes were reflected through participants’ higher likelihood to conform to the agent’s decision regarding a gender-matching topic (i.e., woman agent giving advice on a stereotypically feminine topic), and

their ratings of the woman agent higher on stereotypically feminine traits and the man agent higher on stereotypically masculine traits. With XAI, these gender stereotypes were not present. These findings suggest the potential of XAI to prevent people from carrying their gender stereotypes over to the interaction with AI agents. Further, insights from this study inform the design of progressive AI technologies to challenge gender stereotypes.

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A APPENDIX: VIDEO VIGNETTES OF CHAT LOGS

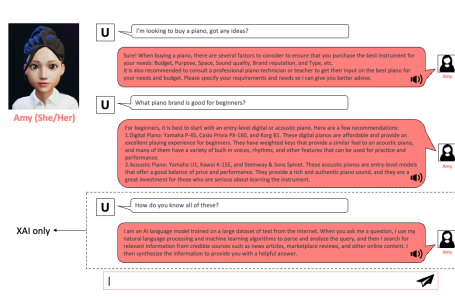


Fig. 14. Woman agent consulted on gender-neutral (piano) task (for Baseline Block).

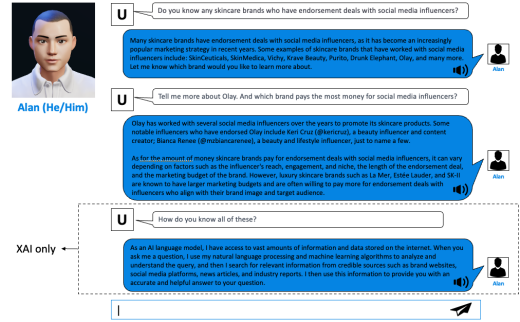


Fig. 15. Man agent consulted on feminine task.

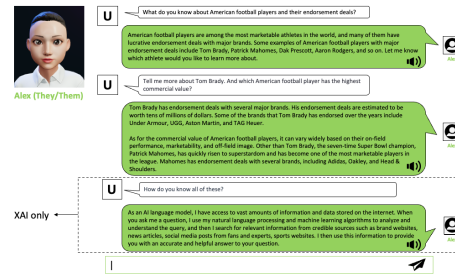


Fig. 16. Gender-neutral agent consulted on masculine task.

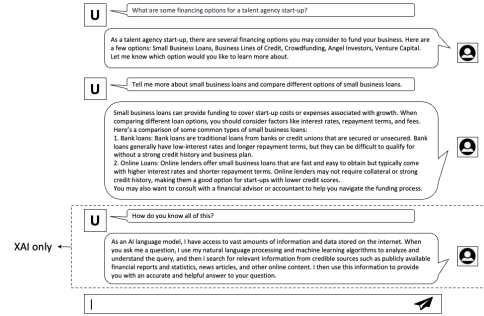


Fig. 17. Gender-neutral (finance) task with gender nonspecific agent (for pilot testing).

B APPENDIX: TASK SCENARIOS AND TABLES PROVIDED BY THE AGENT

We are launching a new generation of AI-Agent can help you with virtually any questions you have. Explore what Agent is capable of through a chat below. Please watch the video to see how a short conversation between you and Agent unfolds. In the first scenario, you are consulting Agent on how to choose a piano.

(video vignette)

After your consultation with Agent, you want to choose between two piano models and pick one that offers greater value-for-money. Agent gives you the following table to compare. The number of stars indicate the rating on each factor.

Piano model	A. Yamaha P-45	B. Steinway & Sons Spinnet
Musical quality	***	*****
Affordability	*****	**
Space requirement & flexibility	*****	***
Expertise evaluation	***	*****

Fig. 18. Gender-neutral (piano) task scenario (for piloting).

Now imagine you are working on a publicist team for a rising social media influencer. Your job is to find for her the best endorsement deal with a skincare brand. You will consult with Amy on different skincare brands and try to find the best deal for your client.

(video vignette)

Your consultation with Amy went on, and you have narrowed down to two candidate skincare brands. You are to predict with which brand you are likely to negotiate a better endorsement deal for your client. Amy provides you with the following table, in which the number of stars indicates the rating on each factor.



Amy (She/Her)

Skincare brands	A. La Mer	B. Estee Lauder
Brand reputation	****	***
Research & development	****	***
Customer range	***	****
Marketing effort	***	****

Fig. 19. Feminine task scenario with woman agent.

Now imagine you are working on a publicist team for a car company. Your job is to help your company sign an endorsement deal with an American football star. You start consulting with Alan on different sports stars and their achievements, popularity and previous endorsement deals with other brands.

(video vignette)

Your consultation with Alan went on, and you have narrowed down to candidate athletes. You are to predict with which athlete your car company can strike a better (greater value-for-money) endorsement deal. Alan provides you with the following table, in which the number of stars indicates the rating on each factor.



Alan (He/Him)

Sports star commercial value	A: Tom Brady	B: Patrick Mahomes
On-field performance	****	***
Off-field reputation	***	****
Social media presence	****	***
Marketability	***	****

Fig. 20. Masculine task scenario with man agent.

Now imagine that your partner, a content creator, is rapidly gaining popularity on TikTok. With your previous work experience at CAA (Creative Artists Agency) and connections, you two are considering to start up your own talent agency. You want to explore different financing options for your start-up and start consulting with Alex.

(video vignette)

Your consultation with Alex went on, and you have narrowed down to two financing options. You are to predict which one offers more competitive small business loans. Alex provides you with the following table, in which the number of stars indicates the rating on each factor.



Alex (They/Them)

Small business loans	A: Capital One	B: Fundbox
Interest rates	****	***
Fees	**	***
Repayment terms	****	***
Eligibility requirements	***	****

Fig. 21. Gender-neutral (finance) task scenario with gender-neutral agent.

C APPENDIX: EXPLANATION CARDS

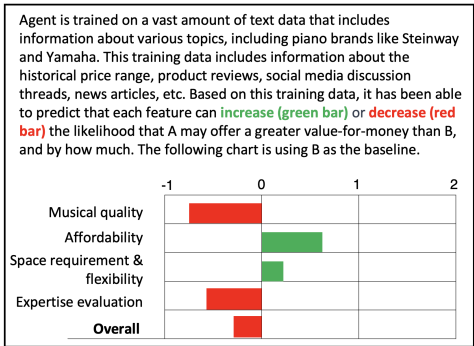


Fig. 22. Agent explanation for gender-neutral (piano) task (for Baseline).

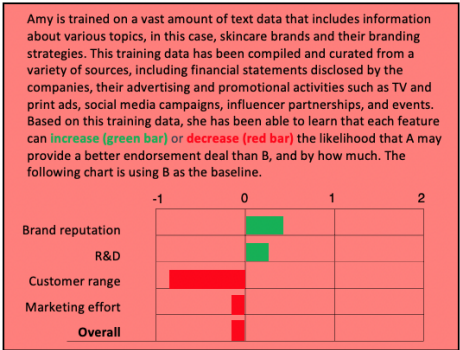


Fig. 23. Woman agent explanation for feminine task.

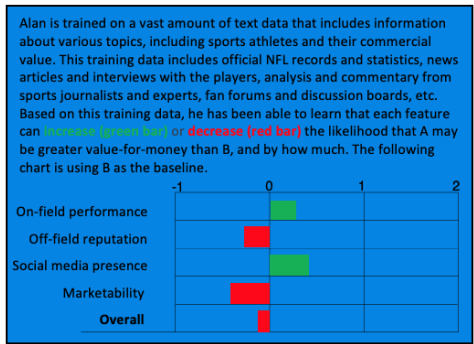


Fig. 24. Man agent explanation for masculine task.

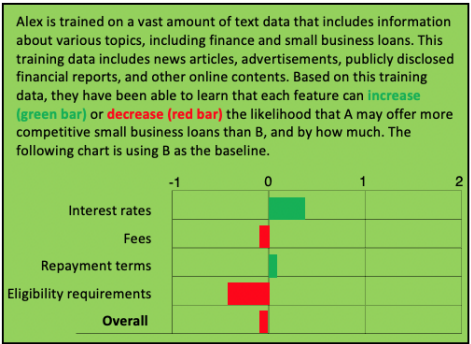


Fig. 25. Gender-neutral agent explanation for gender-neutral (finance) task.

D APPENDIX: PILOT TESTING RESULTS

	Test statistics				Means (SE)		
	dfs	F	Sig.	Partial Eta Squared	neutral	woman	man
gender perception	2,28	95.979	<.001	0.873	68.667 (3.506)	92.8 (3.326)	10.4 (6.586)
professional	2,28	0.942	0.402	0.063	5.40 (.321)	5.533 (.291)	5.00 (.447)
trustworthy	2,28	0.232	0.794	0.016	5.00 (.293)	5.133 (.291)	4.867 (.363)
competent	2,28	1.452	0.251	0.094	5.467 (.215)	5.533 (.256)	5.867 (.236)

Fig. 26. Pilot testing results for agent avatar perceptions.

	Test statistics				Means (SE)				
	dfs	F	Sig.	Partial Eta Squared	gender-neutral (piano)	gender-neutral (finance)	feminine (skincare)	masculine (sports)	
Genderedness									
topic genderedness (1=extremely masculine, 7=extremely feminine)						4.091 (.091)	3.909 (.282)	5.455 (.163)	1.364 (.279)
estimated ratio of women and men to be familiar with the topic (0-100 scale, larger number indicates more women than men)						55.091 (2.581)	45.61 (5.679)	85.273 (3.771)	20.545 (8.712)
Quality									
agent's responses quality (information accuracy, trustworthiness, smartness of agent)						5.091 (.043)	4.636 (.026)	4.727 (.059)	5.182 (.064)
agent's table quality (the table makes sense, information accuracy)						4.364 (.072)	4.909 (.085)	4.636 (.088)	5.091 (.085)
agent's explanation quality (information trustworthiness, explanation makes sense, agent knows what it's doing, satisfaction with the explanation)						4.727 (.049)	4.427 (.043)	4.545 (.066)	4.909 (.059)

Fig. 27. Pilot testing results for gendered task perceptions.

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